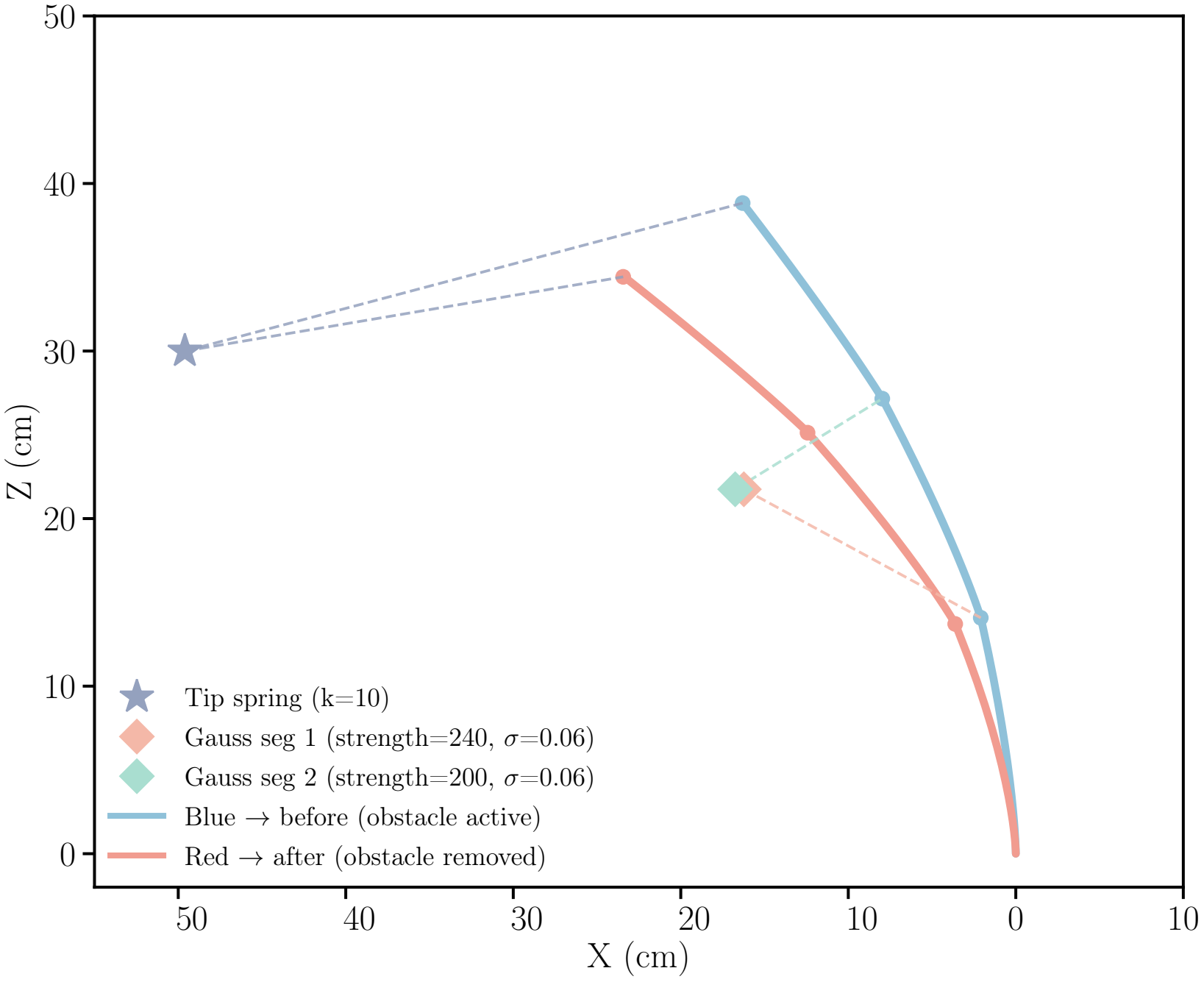

50
40
30
20
10
0
Z (cm)
50
40
30
20
10
0
10
X (cm)
Tip spring (k=10)
Gauss seg 1 (strength=240, σ=0.06)
Gauss seg 2 (strength=200, σ=0.06)
Blue → before (obstacle active)
Red → after (obstacle removed)